Heat Shield – Problem Statement

The problem

Heat is one of the most dangerous hazards for people who work outdoors or in hot places, but it is often ignored. In 2010, 4,190 U.S. workers lost work days because of heat illness, and 40 died (NIOSH, p. 12). The real numbers are likely higher, because many heat illnesses are never recognised or reported. The biggest gap is acclimatization: new workers are often put on full shifts in the heat with no slow start. In OSHA's review of 20 heat cases, this was the safety step most often missing and the one most linked to deaths (NIOSH, p. 12 and p. 33).

Who is affected

Outdoor and hot-indoor workers, such as construction crews, landscapers, roofers, warehouse staff, kitchen workers and farm workers. New workers, and workers returning after a week or more away, are at the highest risk. Crop workers die from heat at about 20 times the rate of all U.S. workers (NIOSH, p. 12). Small employers are also affected, because they often have no safety staff to plan for heat.

Why it isn't fixed yet

There is still no federal heat law. OSHA proposed a rule in August 2024, but it is not final (OSHA rulemaking page). Most states have no heat rule either. For a small employer, writing a heat plan means reading a regulation, a long NIOSH document and a weather forecast at the same time, so many never do it. Some workers also believe myths, such as "if you are still sweating, you cannot get heat stroke" (NIOSH, p. vi).

What Heat Shield will do

Heat Shield will help small employers in Boston make a simple heat safety plan that covers water, rest, shade, a slow start for new workers, and what to do in an emergency, without needing to read long government documents.